

Data Science Course Syllabus

Archer Infotech, Kothrud Pune | Classroom & live-online batches | Updated September 2026

1. Introduction to Data Science

- What is Data Science?
- Evolution and Applications
- Data-Driven Problem Solving
- Data Scientist, ML Engineer, AI Engineer and Research Roles
- Data Science Lifecycle
- Structured, Semi-Structured and Unstructured Data
- Numerical, Categorical, Time-Series, Text and Image Data

2. Python Programming for Data Science

- Python Installation, Jupyter Notebook and Google Colab
- Variables, Data Types, Operators and Strings
- Lists, Tuples, Sets and Dictionaries
- Conditional Statements and Loops
- Functions, Lambda Functions and Recursion
- map(), filter(), reduce(), zip(), enumerate()
- Object-Oriented Programming
- Exception Handling
- File Handling
- Modules, Packages, pip and Virtual Environments

3. NumPy for Scientific Computing

- ndarray and Array Creation
- Shape, Dimensions, Size and Data Types
- Indexing, Slicing, Boolean and Fancy Indexing
- Reshaping, Flattening, Concatenating and Splitting
- Vectorized Numerical Operations
- Statistical Operations
- Broadcasting
- Matrix Operations
- Linear Algebra
- Random Sampling

4. Pandas for Data Science

- Series and DataFrame
- Importing CSV, Excel, JSON, SQL and Text
- Dataset Inspection
- loc, iloc and Boolean Filtering
- Missing Values, Duplicates and Data Type Cleaning
- Data Transformation
- Sorting and Ranking
- Grouping and Aggregation
- concat(), merge(), join()
- Pivoting and Reshaping
- Date-Time Processing and Time-Series Indexing

5. Mathematics for Data Science

Linear Algebra

- Scalars, Vectors, Matrices and Tensors
- Vector and Matrix Operations
- Dot Product and Matrix Multiplication
- Transpose, Determinant, Inverse and Rank
- Eigenvalues and Eigenvectors
- Orthogonality

Calculus

- Functions and Limits
- Differentiation
- Partial Derivatives
- Chain Rule
- Gradients

Optimization

- Objective, Cost and Loss Functions
- Local and Global Minima
- Gradient Descent
- Learning Rate

6. Probability and Statistics

- Population, Sample, Parameter and Statistic
- Mean, Median, Mode, Variance and Standard Deviation
- Percentiles, Quartiles and IQR
- Probability Rules and Conditional Probability
- Bayes' Theorem
- Discrete and Continuous Random Variables
- PMF, PDF and CDF
- Bernoulli, Binomial, Poisson, Uniform, Normal and Exponential Distributions
- Sampling Methods and Sampling Distribution
- Central Limit Theorem
- Standard Error
- Point and Interval Estimation
- Confidence Intervals
- Hypothesis Testing
- P-Value and Significance Level
- Type I and Type II Errors
- Z-Test, T-Test, Chi-Square, ANOVA and F-Test
- Correlation and Covariance

7. Data Visualization

- Visualization Principles
- Matplotlib
- Seaborn
- Plotly
- Line, Bar, Histogram, Scatter, Box and Area Plots
- Heatmaps, Pair Plots and Regression Plots
- Interactive Visualizations

8. Data Cleaning and Preprocessing

- Data Quality
- Missing Data
- Duplicate Records
- Outlier Detection and Treatment
- Label, Ordinal and One-Hot Encoding
- Target Encoding
- Standardization and Normalization
- Robust Scaling
- Log, Box-Cox and Yeo-Johnson Transformations

9. Exploratory Data Analysis

- Data Profiling
- Univariate, Bivariate and Multivariate Analysis
- Correlation Matrix
- Distribution Analysis
- Normality, Skewness and Kurtosis
- Q-Q Plot
- Outlier Analysis

10. Feature Engineering

- Domain-Based Feature Creation
- Numerical and Categorical Features
- Interaction and Polynomial Features
- Date-Time Features
- Binning and Discretization
- Feature Selection
- Tree-Based and Permutation Feature Importance

11. Machine Learning Fundamentals

- Features, Labels, Models, Parameters and Hyperparameters
- Supervised Learning
- Unsupervised Learning
- Semi-Supervised Learning
- Reinforcement Learning
- Train/Validation/Test Splits
- Bias, Variance, Overfitting and Underfitting
- ML Workflow

12. Regression Algorithms

- Simple and Multiple Linear Regression
- Regression Assumptions
- Polynomial Regression
- Ridge, Lasso and Elastic Net
- MAE, MSE, RMSE, R-Squared, Adjusted R-Squared and MAPE

13. Classification Algorithms

- Logistic Regression
- K-Nearest Neighbors

- Naive Bayes
- Decision Trees
- Support Vector Machines
- Confusion Matrix
- Accuracy, Precision, Recall, Specificity and F1
- ROC-AUC and Precision-Recall Curve

14. Ensemble Learning

- Bagging
- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost
- Voting
- Stacking

15. Unsupervised Machine Learning

- Clustering Fundamentals
- K-Means
- Hierarchical Clustering
- DBSCAN
- Gaussian Mixture Models
- Cluster Evaluation

16. Dimensionality Reduction

- Curse of Dimensionality
- Principal Component Analysis
- t-SNE
- UMAP

17. Model Selection and Validation

- K-Fold and Stratified K-Fold Cross-Validation
- Time-Series Cross-Validation
- Grid Search
- Random Search
- Bayesian Optimization
- Optuna
- Learning Curves
- Model Comparison

18. Machine Learning Pipelines

- Scikit-Learn Pipeline
- ColumnTransformer
- FeatureUnion
- Custom Transformers
- Data Leakage
- Reproducibility

19. Time-Series Analysis and Forecasting

- Trend, Seasonality, Cyclic Patterns and Noise
- Additive and Multiplicative Models
- Decomposition
- Stationarity and ADF Test
- ACF and PACF
- Moving Average
- Exponential Smoothing
- Holt and Holt-Winters
- AR, MA, ARMA, ARIMA and SARIMA
- Lag and Rolling Features
- XGBoost Forecasting
- Forecast Evaluation

20. Natural Language Processing

- NLP Fundamentals
- Text Cleaning and Tokenization
- Stop Words, Stemming and Lemmatization
- Bag of Words and N-Grams
- TF-IDF
- Word2Vec, GloVe and FastText
- Text Classification
- Sentiment Analysis
- Spam Detection
- Named Entity Recognition
- Text Similarity

21. Deep Learning Fundamentals

- Artificial Neurons and Perceptrons
- Neural Network Layers
- Weights and Bias
- Sigmoid, Tanh, ReLU, Leaky ReLU and Softmax
- Forward Propagation
- Loss Functions
- Backpropagation
- SGD, Momentum, RMSProp and Adam
- Dropout, Batch Normalization and Early Stopping
- TensorFlow, Keras and PyTorch

22. Convolutional Neural Networks

- Images, Pixels and Channels
- Convolution, Filters and Feature Maps
- Padding, Stride and Pooling
- CNN Training and Image Augmentation
- LeNet, AlexNet, VGG, ResNet and EfficientNet
- Transfer Learning
- Image Classification
- Object Detection
- Image Segmentation

23. Recurrent Neural Networks

- Sequential Data
- RNN Architecture
- Hidden State
- Backpropagation Through Time
- Vanishing and Exploding Gradients
- LSTM
- GRU
- Sequence Classification and Forecasting

24. Attention and Transformers

- Query, Key and Value
- Attention Scores
- Self-Attention
- Scaled Dot-Product Attention
- Multi-Head Attention
- Positional Encoding
- Encoder and Decoder
- BERT
- GPT Architecture Overview
- T5
- Vision Transformer
- Transfer Learning with Transformers

25. Model Explainability

- Interpretability vs Explainability
- Global and Local Explanations
- Feature Importance
- Permutation Importance
- LIME
- SHAP
- Partial Dependence Plots

26. Anomaly Detection

- Point and Contextual Anomalies
- Z-Score and IQR
- Isolation Forest
- Local Outlier Factor
- One-Class SVM
- Fraud and Equipment-Failure Applications

27. Recommendation Systems

- User-Item Matrix
- Content-Based Filtering
- User-Based and Item-Based Collaborative Filtering
- Matrix Factorization
- SVD
- Precision, Recall, MAP and RMSE

28. Model Deployment

- Training vs Inference
- Batch and Real-Time Prediction
- Pickle and Joblib
- REST APIs
- FastAPI
- Streamlit

29. MLOps

- ML Lifecycle and Reproducibility
- Experiment Tracking
- MLflow
- Data and Model Versioning
- DVC Fundamentals
- Model Registry
- Prediction Monitoring
- Data Drift
- Concept Drift
- Performance Degradation

30. Git and GitHub

- Repository, Clone, Add, Commit, Push, Pull
- Branching and Merging
- Pull Requests and Issues
- Data Science Repository Structure
- Portfolio Development

31. Docker for Data Science

- Containers and Images
- Dockerfile
- Containerizing ML Applications
- Docker Compose

32. Cloud Fundamentals

- Compute, Storage, Databases and Networking
- Cloud Model Training and Deployment
- Azure Machine Learning
- Azure Storage
- AWS S3, EC2 and SageMaker
- Google Cloud Storage and Vertex AI

33. Big Data for Data Science

- Big Data Fundamentals
- Apache Spark Architecture
- PySpark
- Spark DataFrames
- Spark SQL
- Spark MLlib
- ML Pipelines with Spark

34. Data Science Projects

- House Price Prediction
- Customer Churn Prediction
- Loan Default Prediction
- Customer Segmentation
- Sales Forecasting
- Fraud Detection
- Sentiment Analysis
- Image Classification
- Recommendation System

35. Capstone Project

- Problem Definition
- Data Collection
- Data Preparation
- EDA
- Feature Engineering
- Baseline and Advanced Models
- Cross-Validation
- Hyperparameter Tuning
- Model Evaluation
- Explainability
- API Development
- Docker and Cloud Deployment
- MLOps
- GitHub Documentation

36. Interview Preparation

- Python
- Mathematics
- Statistics
- Machine Learning
- Deep Learning
- Data Science Case Studies
- Practical Assignments
- Mock Interviews
- Python
- NumPy
- Pandas
- SQL
- Matplotlib
- Seaborn
- Plotly
- SciPy
- Statsmodels
- Scikit-Learn
- XGBoost
- LightGBM
- CatBoost
- TensorFlow
- Keras
- PyTorch
- NLTK

- spaCy
- Hugging Face Transformers
- OpenCV
- SHAP
- LIME
- Optuna
- MLflow
- FastAPI
- Streamlit
- PySpark
- Databricks
- Docker
- Git
- GitHub
- Azure / AWS / GCP
- Jupyter Notebook
- Google Colab

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